PTF ALL

Equipment Cooling Supply Fan Inoperative (Aft) - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace E6 Rack Fan [M3237]
Repair Wiring D14145-D14143
Repair Wiring D14143-Circuit Breaker C4069
Replace EQUIPMENT COOLING E6 RACK FAN POWER Circuit Breaker [C4069]
Repair Wiring D14143-Circuit Breaker C4070
Replace EQUIPMENT COOLING E6 RACK CONTROL Circuit Breaker [C4070]
Replace E6 Rack Fan Control Relay [R1207]
Replace E6 Rack Fan Interrupt Relay [R1205]
Repair Wiring D14143-D14139
Repair Wiring D652-D14145
Replace IFE/PASS SEAT Switch (Flight Deck) [P5-13S3]
Replace E6 Rack Fan Diode [R1208]

A. Initial Evaluation

(1) For the E6 rack fan, M3237, do this task:
Aft Equipment Cooling Fan - Functional Test, AMM TASK 21-27-00-700-808.

(a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.

(b) If the system operates as specified in the AMM task, then there was an intermittent fault.

B. Fault Isolation Procedure

(1) Do this check of the internal components of the E6 rack fan, M3237:

(a) Disconnect connector D14145 from the E6 rack fan, M3237.

(b) Do a check for continuity between these pairs of pins of connector D14145 on the E6 rack fan, M3237 (WDM 21-27-51):

1) Pin 1 and pin 2.

2) Pin 1 and pin 3.

3) Pin 2 and pin 3.

4) Pin 5 and pin 6.

(c) If there is not continuity, then do these steps:

1) Replace the E6 rack fan, M3237. These are the tasks:

- Aft Equipment Cooling Exhaust Fan Removal, AMM TASK 21-27-04-000-802,
- Aft Equipment Cooling Exhaust Fan Installation, AMM TASK 21-27-04-400-802.

2) Do the AMM test of the E6 rack fan that you did at the start of this task.

3) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If there is continuity, then continue.

(2) Do this check of the wiring:

(a) Disconnect connector D14143 from the E6 rack fan control relay, R1207.

(b) Do a wiring check between these pins of connector D14145 at the E6 rack fan and these pins of connector D14143 at the E6 rack fan control relay (WDM 21-27-51):

D14145		**D14143**
pin 1	.....	pin D1
pin 2	.....	pin C1
pin 3	.....	pin B1

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(c) Do a wiring check to make sure that pin 7 of D14145 has continuity to ground.

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D14145 to the E6 rack fan, M3237.

3) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

4) Do the AMM test of the E6 rack fan that you did at the start of this task.

5) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Re-connect connector D14145 to the E6 rack fan, M3237.

(3) Do this check of the voltage at the E6 rack fan control relay:

(a) Do a check for three-phase voltage between pins D2, C2, and B2 of connector D14143 and ground (WDM 21-27-51). The correct voltage is three-phase 115 VAC.

(b) If the correct voltage is not present, then do this check of the wiring:

1) Open the EQUIPMENT COOLING E6 RACK FAN POWER circuit breaker, C4069 [P6-5, D8].

2) Do a wiring check between these pins of connector D14143 at the E6 rack fan control relay and this terminal of circuit breaker C4069 (WDM 21-27-51):

D14143		**C4069**
pin D2	.....	term A1
pin C2	.....	term B1
pin B2	.....	term C1

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

c) Close the EQUIPMENT COOLING E6 RACK FAN POWER circuit breaker, C4069 [P6-5, D8].

d) Do the AMM test of the E6 rack fan that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

b) Close the EQUIPMENT COOLING E6 RACK FAN POWER circuit breaker, C4069 [P6-5, D8].

c) Replace the EQUIPMENT COOLING E6 RACK FAN POWER circuit breaker, C4069.

d) Do the AMM test of the E6 rack fan that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the correct voltage is present, then continue.

(4) Do this check of the voltage at the E6 rack fan control relay:

(a) Do a check for voltage between pin X1 of connector D14143 and ground (WDM 21-27-51). The correct voltage is 28 VDC.

(b) If the correct voltage is not present, then do this check of the wiring:

1) Open the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].

2) Do a wiring check between this pin of connector D14143 at the E6 rack fan control relay and this terminal of circuit breaker C4070 (WDM 21-27-51):

D14143		**C4070**
pin X1	.....	term =L

3) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

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c) Close the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].

d) Do the AMM test of the E6 rack fan that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

4) If you do not find a problem with the wiring, then do these steps:

a) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

b) Close the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].

c) Replace the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070.

d) Do the AMM test of the E6 rack fan that you did at the start of this task.

e) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the correct voltage is present, then continue.

(5) Replace the E6 rack fan control relay, R1207.

(a) Do the AMM test of the E6 rack fan that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(6) Replace the E6 rack fan interrupt relay, R1205.

(a) Do the AMM test of the E6 rack fan that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(7) Do this check of the wiring:

(a) Disconnect connector D14143 from the E6 rack fan control relay, R1207.

(b) Disconnect connector D14139 from the E6 rack fan interrupt relay, R1205.

(c) Do a wiring check between this pin of connector D14143 at the E6 rack fan control relay and this pin of connector D14139 at the E6 rack fan interrupt relay (WDM 21-27-51):

D14143		**D14139**
pin X1	.....	pin 7

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

3) Re-connect connector D14139 to the E6 rack fan interrupt relay, R1205.

4) Do the AMM test of the E6 rack fan that you did at the start of this task.

5) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Re-connect connector D14143 to the E6 rack fan control relay, R1207.

(g) Re-connect connector D14139 to the E6 rack fan interrupt relay, R1205.

(8) Do this check of the wiring:

(a) Disconnect connector D652 from the IFE/PASS SEAT switch (flight deck), P5-13S3.

(b) Disconnect connector D14145 from the E6 rack fan, M3237.

(c) Do a wiring check between this pin of connector D652 at the IFE/PASS SEAT switch (flight deck) and this pin of connector D14145 at the E6 rack fan (WDM 21-27-51, WDM 24-28-41):

D652		**D14145**
pin 7	.....	pin 5

(d) Do a wiring check to make sure that pin 6 of D652 has continuity to ground.

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D652 to the IFE/PASS SEAT switch (flight deck), P5-13S3.

3) Re-connect connector D14145 to the E6 rack fan, M3237.

4) Do the AMM test of the E6 rack fan that you did at the start of this task.

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5) If the system operates as specified in the AMM task, then you corrected the fault.

(f) If you do not find a problem with the wiring, then continue.

(g) Re-connect connector D14145 to the E6 rack fan, M3237.

(9) Do this check of the continuity through the IFE/PASS SEAT switch (flight deck) (WDM 21-27-51):

(a) For each switch position below, set the switch to the specified position and do a check for continuity between the specified pins on the switch:

1) Switch position OFF, make sure there is NOT continuity between pin 6 and pin 7.

2) Switch position ON, make sure there is continuity between pin 6 and pin 7.

(b) If there was a problem with the continuity, then do these steps:

1) Replace the IFE/PASS SEAT switch (flight deck), P5-13S3.

2) Re-connect connector D652 to the IFE/PASS SEAT switch (flight deck), P5-13S3.

3) Do the AMM test of the E6 rack fan that you did at the start of this task.

4) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If there was not a problem with the continuity, then continue.

(d) Re-connect connector D652 to the IFE/PASS SEAT switch (flight deck), P5-13S3.

(10) Do this check of the E6 rack fan diode:

(a) Disconnect the wires from the power terminals of the E6 rack fan diode, R1208 (WDM 21-27-51).

(b) Connect an ohmmeter to do this diode check in the forward direction:

1) Connect the positive probe of the meter to terminal A11 of the E6 rack fan diode.

2) Connect the negative probe of the meter to terminal B11 of the E6 rack fan diode.

3) There will be continuity between the pins if the forward direction is good.

(c) Connect the ohmmeter to do this diode check in the reverse direction:

1) Connect the negative probe of the meter to terminal A11 of the E6 rack fan diode.

2) Connect the positive probe of the meter to terminal B11 of the E6 rack fan diode.

3) There will be no continuity between the pins if the reverse direction is good.

(d) If the diode check was not good in the forward direction or the reverse direction, then do these steps:

1) Replace the E6 rack fan diode, R1208.

2) Do the AMM test of the E6 rack fan that you did at the start of this task.

3) If the system operates as specified in the AMM task, then you corrected the fault.

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